

Lithuania

Lithuanian Competition Council

One of the strategic goals of the Lithuanian Competition Council (LCC) is to achieve a high level of digital maturity. In an era of increasingly complex markets and data-driven business practices, a high level of digital maturity is essential for ensuring effective and timely competition oversight. In pursuit of this objective, the LCC continues to expand its use of computational tools and advanced analytics to modernize enforcement and strengthen its competition oversight.

I. Real-Time Market Surveillance via Web scraping

The LCC uses third party web-based data collection (web scraping) tools for advanced surveillance of selected online markets. These tools allow the authority to collect digital evidence and improve the data used in ongoing investigations and build ex-officio investigations. Use of web scraping allowed the authority to open several investigations in recent years.

II. Streamlining Investigations with Automated Transcription of Audio and Video Evidence

The LCC has implemented automated audio and video transcription tools. These significantly reduce the time and cost of manual transcription and improve the ability to process large volumes of audio and video material during investigations.

III. Participation in the DATACROS Project

As a partner in the DATACROS project, the LCC contributed to the development of a prototype tool assessing corruption risk factors in company ownership structures.¹² The tool is designed to detect collusion, corruption, and money laundering risks across the EU by analyzing firm-level data, with particular focus on complex corporate networks and hidden beneficial ownership.

¹² Transcrime, DATACROS II: Empowering a Tool to Assess Corruption Risk Factors in Firms' Ownership Structure ("DATACROS II") (Feb. 2022-Jan. 2024), Transcrime - Research Centre on Innovation and Crime, <https://www.transcrime.it/en/projects/datacros-ii/>.

IV. Piloting AI Models for Cartel Detection (eDiscovery)

In its effort to modernize investigative capabilities, the LCC has piloted the use of AI-driven eDiscovery tools for cartel detection.

One such tool is an AI-enhanced document review platform specifically designed to detect signs of collusion and corruption. The tool has several built-in cartel detection models: price fixing, market sharing, bid-rigging, information exchange. It functions similarly to standard document review platforms like Nuix, but with an AI engine capable of contextual, emotional and behavioral analysis. It supports over 100 languages and can detect unusual communication patterns or signals of coordination with significantly greater speed and breadth than manual review.

The LCC has conducted several rounds of testing with different sample cases. The results were mixed but promising. In two complex cases with fragmented evidence, the AI significantly outperformed expectations by retrieving a vast majority of the manually identified evidence in just one week—a task that would have taken the case team several months to complete. However, in other cases, the AI underperformed initially until given example documents to learn from, highlighting the importance of contextual training.

The LCC's pilot of AI-driven eDiscovery tools marks a significant step toward more efficient evidence handling in complex cartel investigations. While results varied across cases, the pilot underscored two key insights: (1) AI models can rapidly surface critical evidence when trained on context-specific examples, and (2) human-guided calibration remains essential for effective deployment. Based on encouraging results from initial pilots, the LCC will continue to explore the potential use of this tool in future investigations, especially in cases involving large volumes of unstructured data and limited initial visibility.

V. Data-Driven Screening for Bid-Rigging

The Lithuanian Competition Council has developed data-driven screening tools to identify potential bid-rigging behavior in public procurement processes. These tools are designed to detect red flags and anomalies suggestive of collusive practices—such as coordinated pricing, bid rotation, or unusual bidding patterns—by analyzing structured procurement data.

The screening framework aims to enable more proactive detection of cartels and reduce reliance on *ex post* complaints or whistleblower reports. However, the practical deployment of these tools has faced limitations. Specifically, while the consistency and granularity of publicly available procurement data were initially sufficient, a noticeable decline in quality and detail over time has significantly hindered the effectiveness of algorithmic analysis. Currently, many datasets lack essential variables (e.g., pricing information) which are critical for drawing reliable conclusions. As a result, the LCC has temporarily paused implementation of the screening system.

This experience highlighted a broader need for harmonised, high-quality procurement datasets. The LCC is now using these insights to advocate for improved data governance and to explore inter-agency data-sharing protocols that could support future system reactivation.

VI. Future Plans: Virtual AI Assistants for Institutional Knowledge Management

Looking ahead, the LCC is planning to develop and implement AI-based virtual assistants to improve internal knowledge management. These intelligent agents will assist staff in efficiently retrieving and reviewing institutional information, including legal acts, previous decisions, guidelines, and policy documents. The goal is to reduce time spent on manual searches and enhance decision-making by ensuring faster access to relevant internal resources.